

AugmentedDataset

vpo/augment.py

rewrites the prompt: "give m answers"

m -answer prompt

Policy π_θ

one autoregressive chain $\rightarrow m$ answers

m answers in one chain

Task scoring

tasks/<task>.py · score_multi

via the vpo/reward.py dispatcher

sub_scores: $(m \times k)$ matrix

Patched veRL dispatch $\rightarrow$ vpo_advantage

verl/.../ray_trainer.py · vpo/utils/vpo.py

$E_w[\max_s w \cdot r_s]$, group z-norm

advantages

Standard PPO update

unchanged loss / KL / optimizer / rollout engine